PDN. With DC analysis, we ensure design accounts for the following aspects:

- Steady state voltage drop at different system level components from respective voltage source.
- Current densities along the layout and its uniform distribution throughout.
- Maximum current flowing through individual vias on PCB.
- Power loss contribution of each individual board layer in case of multilayer PCB stack up.

Simulation setups for running DC analysis are fast and accurate. DC analysis is the first step before doing any kind of frequency domain or time domain analysis of the PDN.

Fig. 4 highlights a typical failure case of PDN that can be analysed and fixed through DC analysis. It shows a power layout (highlighted in green) from VR output to the load point. At load point, the voltage drops from 1V to 0.74V, which will render the chip non-functional. With such gaps identified, the layout can be optimised to reduce voltage drops and meet design specs for all system level components by ensuring path resistance is less than target resistance R_{Target}.

Fig. 5 highlights another failure case of a PDN with ~6.5A current flowing through single via in the design. Typical specification for a standard PCB design is maximum 2A current though via. If not accounted for, such high current would lead to via burn on PCB. Such issues can be fixed by either adding more vias in the regions with high current densities or changing via location/pattern in the layout.

Frequency domain analysis. Frequency domain analysis is helpful to understand PDN performance across a range of operational frequencies. With FDA, PDN impedance Z as a function of frequency is plotted and compared against target impedance Z_{Target} to ensure $Z < Z_{Target}$ for system range of operational frequencies.

It is also helpful to understand power delivery performance sensitivity with different type of capacitors, such as bulk caps, motherboard caps, land side capacitor, etc (as shown in Fig. 2), number of capacitors and their respective placements. With optimal type, combination, and location of the capacitors studied and worked out through FDA, we can optimise and design a cost-effective power delivery solution, which is based on actual analysis of the system and not just thumb rules or generic vendor guidelines.

Fig. 6 displays an impedance profile of a typical PDN. The various crests and troughs in the impedance profile are result of resonance and anti-resonance between various series and parallel resistor-inductor-capacitor (RLC) combinations, as shown in Fig. 3. With reference to PDN shown in Fig. 3, the constant/flat impedance region in low frequency range of the profile is governed by effective resistance between the silicon die and VR source.

The first crest in the profile, around 100kHz frequency range, is dictated by parasitic inductance of the bulk capacitor. This is also re-

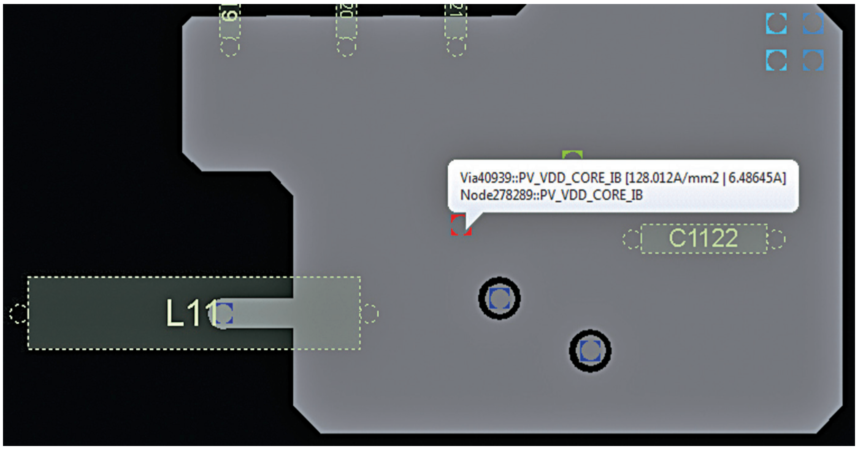

Fig. 5: PDN failure case with high current through individual via

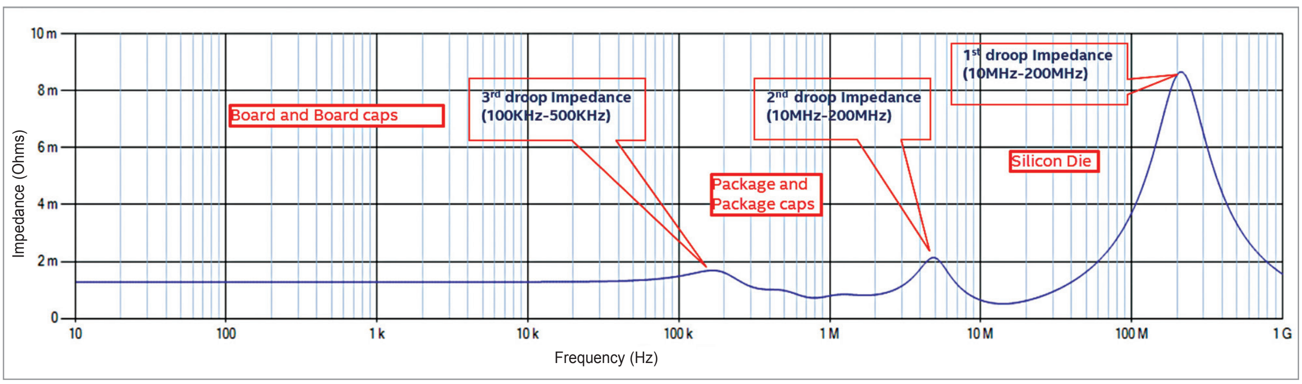

Fig. 6: Impedance as a function of frequency for a typical PDN network